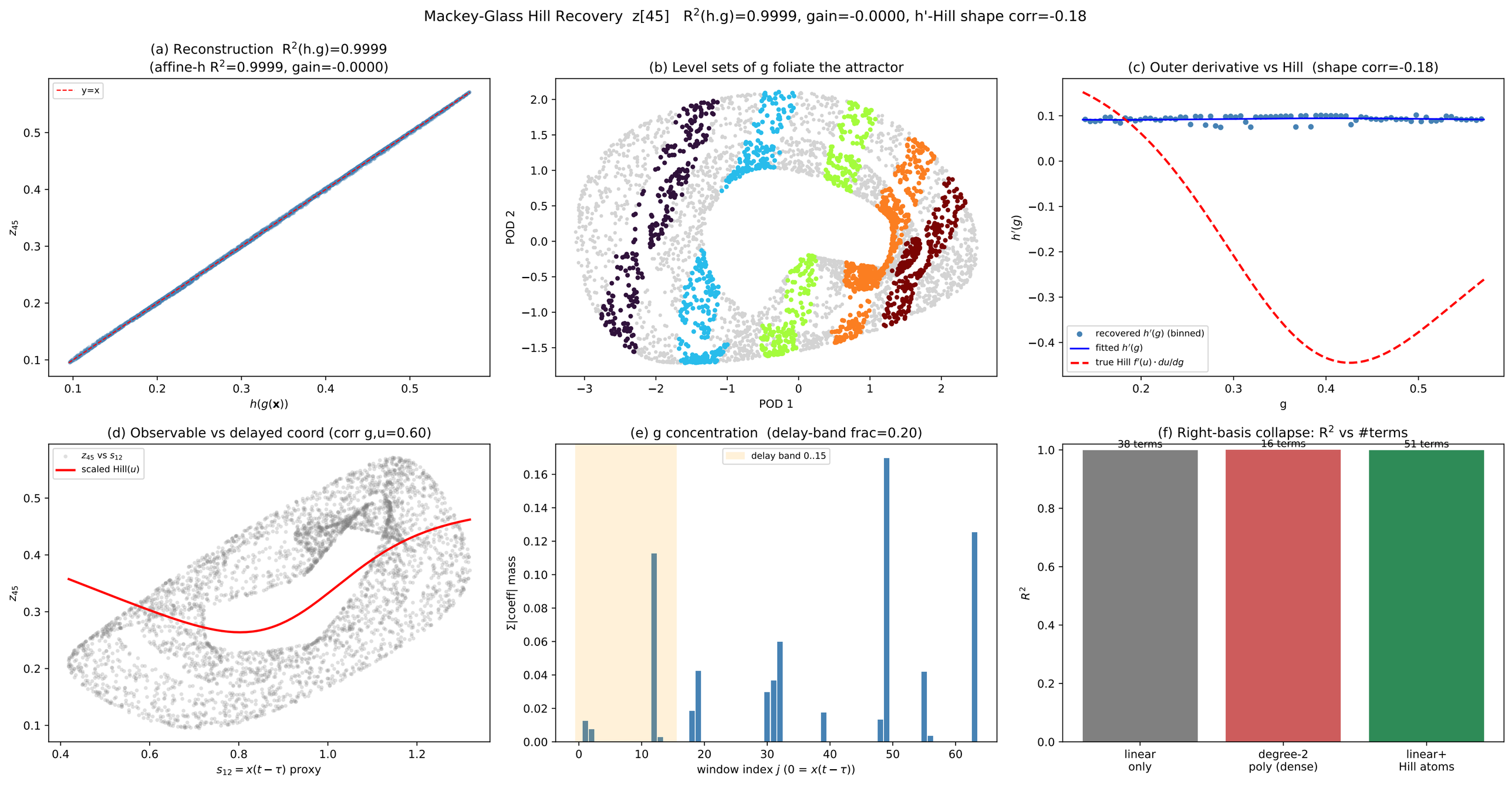

Mackey-Glass Hill Recovery z[45] R²(h.g)=0.9999, gain=-0.0000, h'-Hill shape corr=-0.18
(a) Reconstruction R²(h.g)=0.9999
(affine-h R²=0.9999, gain=-0.0000)
y=x
z45
h(g(x))
(b) Level sets of g foliate the attractor
POD 2
POD 1
(c) Outer derivative vs Hill (shape corr=-0.18)
h'(g)
g
recovered h'(g) (binned)
fitted h'(g)
true Hill f'(u)·du/dg
(d) Observable vs delayed coord (corr g,u=0.60)
z45 vs s12
scaled Hill(u)
z45
s12 = x(t − τ) proxy
(e) g concentration (delay-band frac=0.20)
delay band 0..15
Σ|coeff| mass
window index j (0 = x(t − τ))
(f) Right-basis collapse: R² vs #terms
38 terms
16 terms
51 terms
R²
linear
only
degree-2
poly (dense)
linear+
Hill atoms